National Research University Higher School of Economics (HSE)

Curriculum

Field of study 01.03.02 Applied Mathematics and Information Science

Educational Programme "Data Science and Business Analytics"

Specializations: "Data Science in Applied Research", "Data Science in Business", "Data Science in Finance"

Implementing unit: Faculty of Computer Science, HSE - Moscow

3 rd, 2026/2027 academic year

APPROVED

24.04.2026

Vice Rector

ROSHCHIN S.Y.

Signed with EDS

Length of Programme: 4 years

Years of Study: 2024/2025 - 2027/2028

Mode of Study: Full Time

Degree: Bachelor’s degree

Block Code	Course	Subject type	Department	Period of execution	Credits	Total Academic Hours	Contact Hours	Allocation of Contact Hours				Additional Information
								1	2	3	4	
	Degree Progamme				**60,00**	**2 280**	**798**	**220**	**204**	**260**	**162**	
	Minor				**10,00**	**380**	**124**	**28**	**28**	**40**	**28**	
1	Minor	E			10,00	380	124	28	28A	40	28A	Foreign language
	General Courses				**3,00**	**114**	**40**			**40**		
	Optional General Courses				**3,00**	**114**	**40**			**40**		
1	Introduction to law	C	Department of Big Data and Information Retrieval		3,00	114	40			40A		Foreign language
	Data Culture				**5,00**	**190**	**60**	**28**	**30**		**2**	
1	Databases	C	Department of Software Engineering		4,00	152	56	28	28A			Online Course, Foreign language
2	External Examinations on Digital Skills. Final Level	C	отдел развития цифровых компетенций				2		2A			Foreign language
3	Data Science, AI and Generative Models Independent Test. Advanced Level	C	отдел развития цифровых компетенций		1,00	38	2				2A	Online Course, Foreign language
	Internship				**8,00**	**304**	**4**	**2**			**2**	
	Project Internship				**5,00**	**190**	**2**				**2**	
1	Programming Project 2	C		F	5,00	190	2				2A	Foreign language
	Professional Internship				**3,00**	**114**	**2**	**2**				
1	Work Experience Internship	C		F	3,00	114						Foreign language
2	Educational Internship	C		F			2	2A				Foreign language
	Major				**34,00**	**1 292**	**570**	**162**	**146**	**180**	**130**	
	Core Professional Block (Major)				**5,00**	**190**	**56**	**28**	**28**			
	Core Professional Block (Major)				**5,00**	**190**	**56**	**28**	**28**			
1	Optimization Methods	C	Department of Big Data and Information Retrieval		5,00	190	56	28	28A			Online Course, Foreign language
	Elective Professional Block (Major)				**29,00**	**1 102**	**514**	**134**	**118**	**180**	**130**	
	Elective Courses				**5,00**	**190**	**68**	**28**		**60**	**28**	
	Elective Courses (Block 1)				**5,00**	**190**	**68**			**40**	**28**	

1	Operations Research and Game Theory	E	Department of Big Data and Information Retrieval		5,00	190	68			40	28A	Foreign language
2	Advanced Statistical Methods	E	Department of Big Data and Information Retrieval		5,00	190	102			60	42A	Foreign language
	Compulsory Courses				**12,00**	**456**	**244**	**60**	**64**	**60**	**60**	
1	Time Series Analysis	C	Department of Big Data and Information Retrieval		4,00	152	120			60	60A	Foreign language
2	Deep Learning	C	Department of Big Data and Information Retrieval		4,00	152	68	32	36A			Foreign language
3	Stochastic processes and applications	C	Department of Big Data and Information Retrieval		4,00	152	56	28	28A			Foreign language
	Specializations				**12,00**	**456**	**202**	**46**	**54**	**60**	**42**	
	Educational track "Data Science in Business"				**12,00**	**456**	**204**	**42**	**42**	**66**	**54**	
1	Research Seminar "Data Analysis in Business"	C	Department of Big Data and Information Retrieval		4,00	152	68	14	14	26	14A	Foreign language
2	Investment Management for Business	C	Department of Big Data and Information Retrieval		4,00	152	80			40	40A	Foreign language
3	Product Development: from Idea to Production	C	Department of Big Data and Information Retrieval		4,00	152	56	28	28A			Foreign language
	Educational Track "Data Analysis in Applied Research"				**12,00**	**456**	**188**	**42**	**42**	**60**	**44**	
1	Methods of Mathematic Modeling	C	Department of Big Data and Information Retrieval		4,00	152	56	28	28A			Foreign language
2	Research Seminar "Data Analysis in Applied Research"	C	Department of Data Analysis and Artificial Intelligence		4,00	152	64	14	14	20	16A	Foreign language
3	Applied Statistics for Machine Learning	C	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
	Educational track "Data Science in Finance"				**12,00**	**456**	**186**	**42**	**42**	**60**	**42**	
1	Research Seminar "Data Analysis in Finance"	C	Department of Big Data and Information Retrieval		4,00	152	62	14	14	20	14A	Foreign language
2	Risk management in a bank	C	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
3	Financial Mathematics	C	Department of Big Data and Information Retrieval		4,00	152	56	28	28A			Foreign language

Curriculum agreed:

Academic Supervisor	Voznesenskaia T.V.	17.03.2026
Dean	ARZHANTSEV I.V.	17.03.2026
Head of Centre for Educational Model Design	LEPESHKIN I.A.	23.04.2026

* Subject type:

Compulsory course	C
Elective course	E

Period of execution:

Fixed

F